Project Title

Resume–Job Matching & Talent-Marketplace Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

X[l

Submitted by

D.Shamika Hasini– 2520080033

M.Priyavallika - 2520090183

Team Number:

Section: 10

Under the guidance of

K.Chandusha (Asst.Proffeser)



Hyderabad-500090, Telangana, India.

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY/ARTIFICIAL INTELLIGENCE & DATA SCIENCE



DECLARATION

The Mini Project Report entitled **Resume–Job Matching & Talent-Marketplace Engine** is a record of Bonafide work of **D.Shamika Hasini– 2520080033 , M.Priyavallika - 2520090183** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering/Computer Science and Information Technology/ Artifical Intelligence & data Science to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

D.Shamika Hasini– 2520080033

M.Priyavallika - 2520090183

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



CERTIFICATE

This is to certify that the mini project based report entitled **Resume–Job Matching & Talent-Marketplace Engine** is a bonafide work done and submitted by **D.Shamika Hasini–2520080033 , M.Priyavallika - 2520090183** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering/ Computer Science and Information Technology/Artificial Intelligence & Data Science, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, K.Chandusha which cannot be expressed in words for his/her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. P.Venkateshwar Rao (CSE)/Dr. Khaja Shareef (CSIT)/ Dr. Sreenath Itikela (AI&DS)**, Head of the *Department for Computer Science Engineering/ Department for Computer Science and Information Technology/Department of Artificial Intelligence and Data Science* for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

D.Shamika Hasini– 2520080033

M.Priyavallika - 2520090183

TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	6
2.	Introduction	7
3.	Literature survey	8
4.	Problem Statement and Objectives	9
5.	Hardware and Software requirements	10
6.	Implementation	11-12
7.	Results & Discussion	
8.	Conclusion & Future Scope	
9.	References	

1.ABSTRACT

The **Resume–Job Matching & Talent-Marketplace Engine** is a data-structures-driven platform designed to simplify the process of connecting candidates with suitable job opportunities based on their skills, resume information, and career interests. The system analyzes uploaded resumes, extracts relevant skills, identifies skill gaps, and matches candidates with appropriate job requirements. It uses data structures and algorithms such as **HashMap/HashSet for efficient skill lookup, N-gram scanning for skill identification, Min-Heap for Top-K job ranking, Graph-based algorithms for learning roadmap generation, and optimization techniques for candidate-job assignment**. The platform also provides career recommendations and learning roadmaps to help candidates improve their skills according to their target roles. In addition, the talent marketplace enables interaction between candidates and job opportunities through a structured matching process. The system is implemented using a modern web-based architecture with **React and Vite for the frontend, Node.js and Express for the backend, and MySQL for data management**. The project demonstrates how practical Data Structures and Algorithms can be integrated into a real-world recruitment application to improve search, ranking, matching, and recommendation processes.

Team Name:

Team Members:

1. D.Shamika Hasini , 2520080033
2. M.Priyavallika , 2520090183

2. INTRODUCTION

The **Resume–Job Matching & Talent-Marketplace Engine** is a data-structures-and-algorithms-based platform developed to improve the process of connecting candidates with suitable job opportunities. Traditional recruitment processes often require candidates and recruiters to manually search through large amounts of resume and job information, making it difficult to identify the most relevant matches efficiently. This project addresses this challenge by analyzing candidate resumes, extracting relevant skills, comparing them with job requirements, and ranking suitable opportunities.

The system uses **Data Structures and Algorithms** to perform important recruitment operations efficiently. Hash-based structures are used for fast skill lookup and matching, N-gram scanning supports skill identification, and a **Min-Heap** is used to obtain the Top-K relevant job matches. Graph-based algorithms support career and learning-roadmap generation, while optimization techniques are used for candidate-job assignment. These algorithms allow the platform to process and organize recruitment information systematically.

The platform also provides **skill-gap analysis, career recommendations, learning roadmaps, and talent-marketplace functionality**, helping candidates understand missing skills and identify suitable career opportunities. The project demonstrates how DSA concepts can be applied to a practical recruitment problem to improve search, ranking, matching, and recommendation operations.

3. LITERATURE SURVEY

3.1 LinkedIn Talent Solutions

LinkedIn Talent Solutions provides recruitment and candidate-matching capabilities. Its system intelligently matches open roles with relevant members and allows recruiters to filter candidates using skills, location, and seniority. It also provides recommendations for potential candidates who best match job criteria. These features are related to our project because **Resume–Job Matching & Talent-Marketplace Engine** also focuses on skill-based candidate-job matching and ranking.

3.2 Indeed Smart Sourcing

Indeed Smart Sourcing uses candidate profiles, resumes, skills, experience, and preferences to identify suitable candidates for job requirements. Its AI-powered matching connects job posts with relevant job seekers and helps recruiters reduce manual resume screening. This provides a useful reference for our project's resume analysis, candidate matching, and talent-marketplace functionality.

3.3 Workday Candidate Skills Match

Workday Candidate Skills Match uses machine-learning algorithms to calculate the similarity between skills extracted from a candidate's application/resume and the requirements of a job requisition. It provides match scores and identifies required, relevant, and missing skills. This approach is closely related to our project's **skill extraction, skill-gap analysis, and resume-job matching** modules.

3.4 Eightfold Talent Intelligence

Eightfold Talent Intelligence is an AI-based talent platform that matches candidates with jobs using skills, capabilities, potential, and work-related information. It goes beyond simple resume keyword matching and supports talent acquisition, career development, and skill-gap identification. Our project follows a similar overall concept while implementing the core matching and recommendation operations using practical **Data Structures and Algorithms**.

3.5 Eightfold AI Talent Matching

Eightfold AI Talent Matching describes a talent-matching engine that combines information extracted from unstructured candidate data with structured features to produce explainable match scores. The system aims to move beyond simple keyword matching and provide deeper candidate-job relationships. This influenced the design direction of our project, particularly the focus on **skill-based matching and explaining why candidates or jobs are recommended**.

4. Problem Statement and Objectives

4.1 Problem Statement

To develop an efficient DSA-based platform that automatically matches candidates with suitable job opportunities by analyzing resume skills, job requirements, and skill gaps.

4.2 Problem Description

Traditional recruitment requires candidates and recruiters to manually search and compare large amounts of resume and job information. This makes it difficult to identify relevant opportunities efficiently. The proposed system addresses this challenge through automated **skill extraction, resume–job matching, ranking, skill-gap analysis, and talent recommendations**.

4.3 Objectives

The main objectives of the **Resume–Job Matching & Talent-Marketplace Engine** are:

1. **Automate Resume Analysis**
   Extract relevant skills and information from uploaded resumes.
2. **Perform Resume–Job Matching**
   Compare candidate skills with job requirements and identify suitable opportunities.
3. **Provide Top Job Recommendations**
   Rank and retrieve the most relevant jobs using efficient data structures and algorithms.
4. **Identify Skill Gaps**
   Determine missing or insufficient skills required for a candidate's target jobs.
5. **Generate Career and Learning Recommendations**
   Recommend suitable career paths and learning roadmaps based on identified skill gaps.
6. **Provide a Talent-Marketplace Platform**
   Connect candidates and job opportunities through a centralized recruitment and matching system.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- **Processor:** Intel Core i3 / equivalent or above
- **RAM:** Minimum 8 GB
- **Storage:** Minimum 10 GB free space
- **Display:** 1366 × 768 resolution or above
- **Internet:** Required for accessing the web application and related services

Software Requirements:

- **Operating System:** Windows / Linux / macOS
- **Frontend:** React.js with Vite
- **Backend:** Node.js and Express.js
- **Database:** MySQL
- **Programming Languages:** JavaScript, HTML, CSS
- **Development Environment:** Visual Studio Code
- **Web Browser:** Google Chrome / Microsoft Edge / Mozilla Firefox
- **Version Control:** Git / GitHub
- **Database Management:** MySQL Workbench

6. IMPLEMENTATION

6.1 Resume Processing and Skill Extraction

The system allows a user to upload a resume in PDF format. The backend processes the uploaded file and extracts its text using the PDF processing module. The extracted resume text is then passed to the DSA module for skill identification.

The `extractSkills()` function converts the resume text into lowercase tokens and performs **1-gram, 2-gram, and 3-gram scanning**. A `Map` containing predefined skills and aliases is used for efficient skill lookup, while a `Set` stores the identified skills without duplication.

DSA used:

- HashMap / Map
- HashSet / Set
- N-gram scanning

6.2 Resume–Job Matching and Top-K Recommendation

After extracting the candidate's skills, the system retrieves available jobs from the MySQL database. For every job, the candidate's skills are compared with the required skills.

The `pct()` function calculates the percentage of required job skills that are present in the candidate's resume. The system also identifies both **matched skills and missing skills**. The `topK()` function uses a manually implemented **Min-Heap** to efficiently maintain the most relevant job recommendations.

The database currently contains job profiles such as **Data Analyst, ML Engineer, Full Stack Developer, Backend Developer, and NLP Engineer**, each associated with required skills.

DSA used:

- HashSet
- Skill lookup
- Min-Heap
- Top-K selection

6.3 Skill-Gap Analysis and Learning Roadmap

The system identifies the skills missing from a candidate's profile for a selected job. It then uses the prerequisite relationships stored in the `skill_prereqs` MySQL table to determine which additional skills need to be learned.

The `roadmap()` function represents these relationships as a **directed graph**. It uses **DFS** to discover prerequisite skills and **Kahn's Topological Sort** to arrange the skills in a suitable learning order. The generated roadmap provides sequential learning steps with starting and ending weeks and identifies whether a skill is a required skill or a prerequisite.

DSA used:

- Graph
- DFS
- Topological Sort
- HashMap / Map
- HashSet / Set

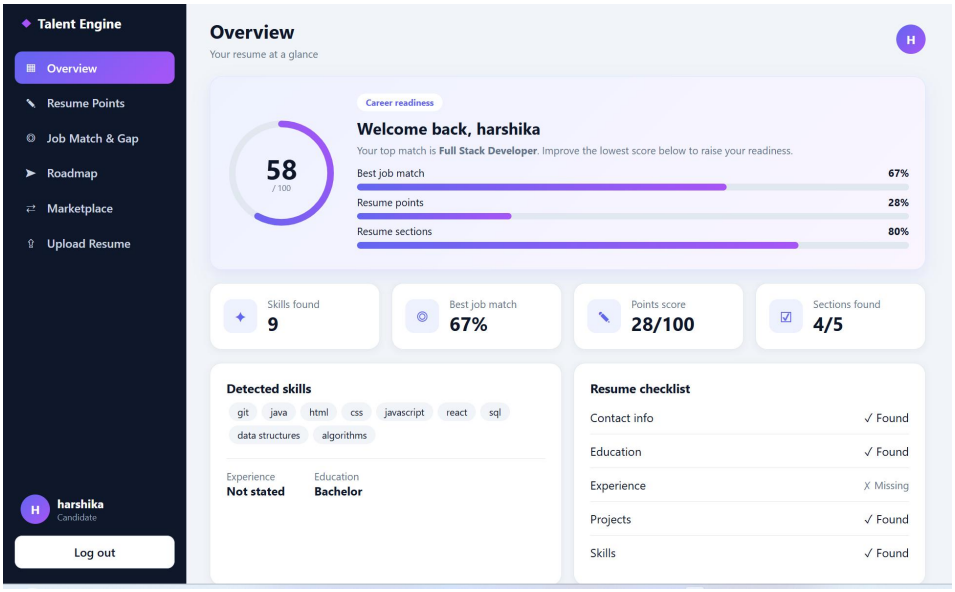
6.4 Talent Marketplace and Candidate–Job Assignment

The marketplace module retrieves candidates and available jobs from the database and extracts the skills of each candidate. The system can determine suitable candidates for a selected job using skill coverage.

The project also implements a **Hungarian Algorithm** for candidate–job assignment. A cost matrix is created using the difference between 100 and the candidate's skill-match percentage. The Hungarian Algorithm then finds an efficient assignment between candidates and jobs. In addition, the `minTeam()` function implements a **Greedy Set Cover approach** to select candidates who collectively cover the required skills of a job.

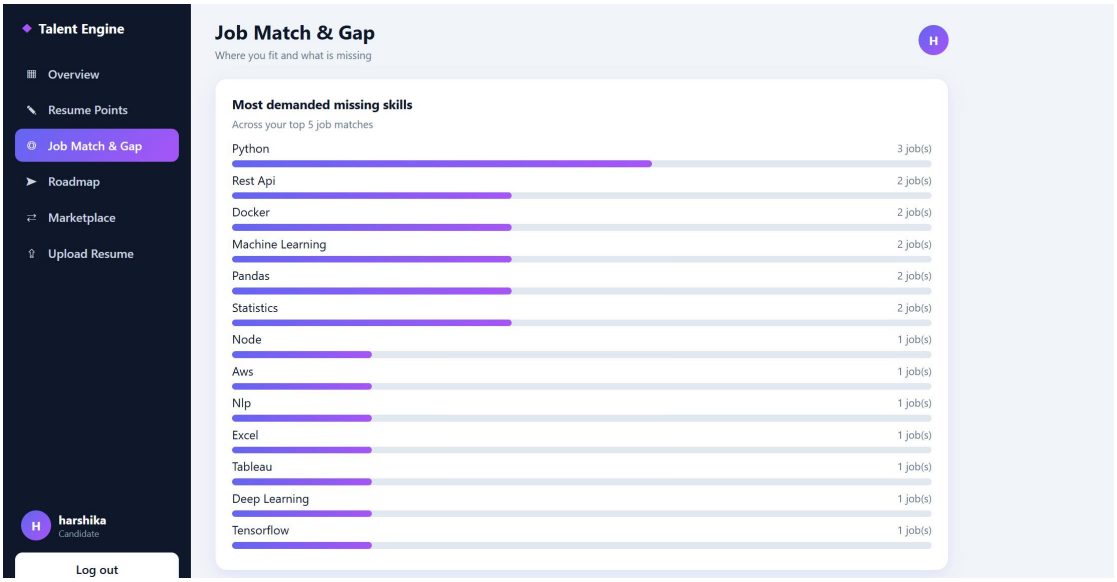
7. RESULTS & DISCUSSION

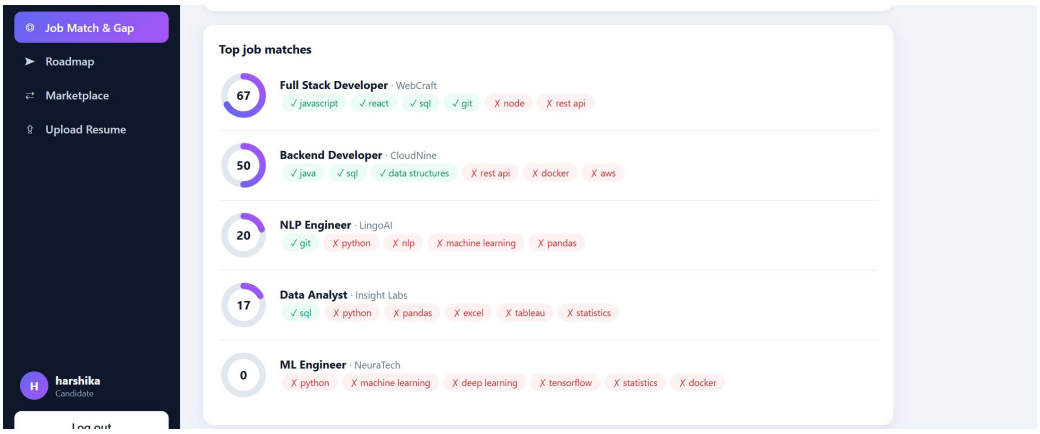
7.1 Resume Upload and Skill Extraction



The system accepts a candidate resume and extracts relevant skills from the uploaded resume. The extracted skills are used as the input for subsequent job matching and skill-gap analysis.

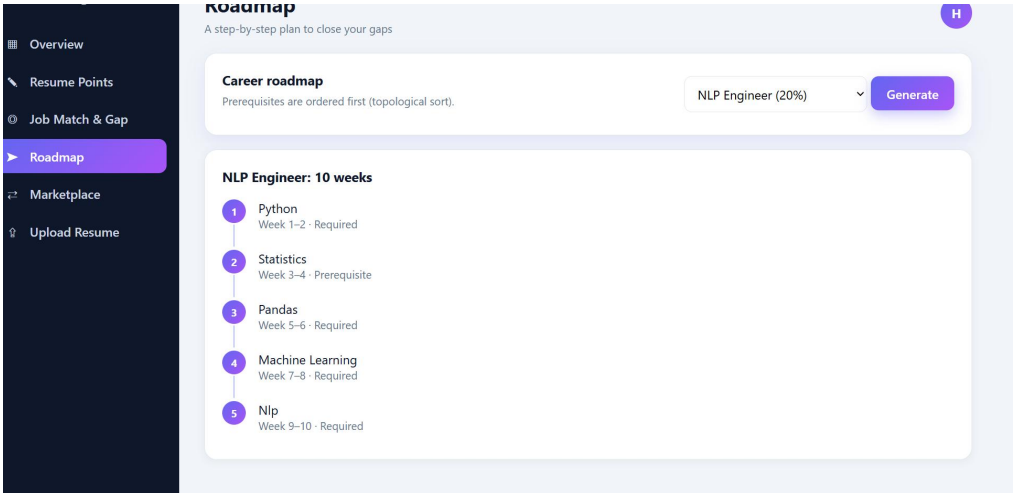
7.2 Job Matching and Recommendations





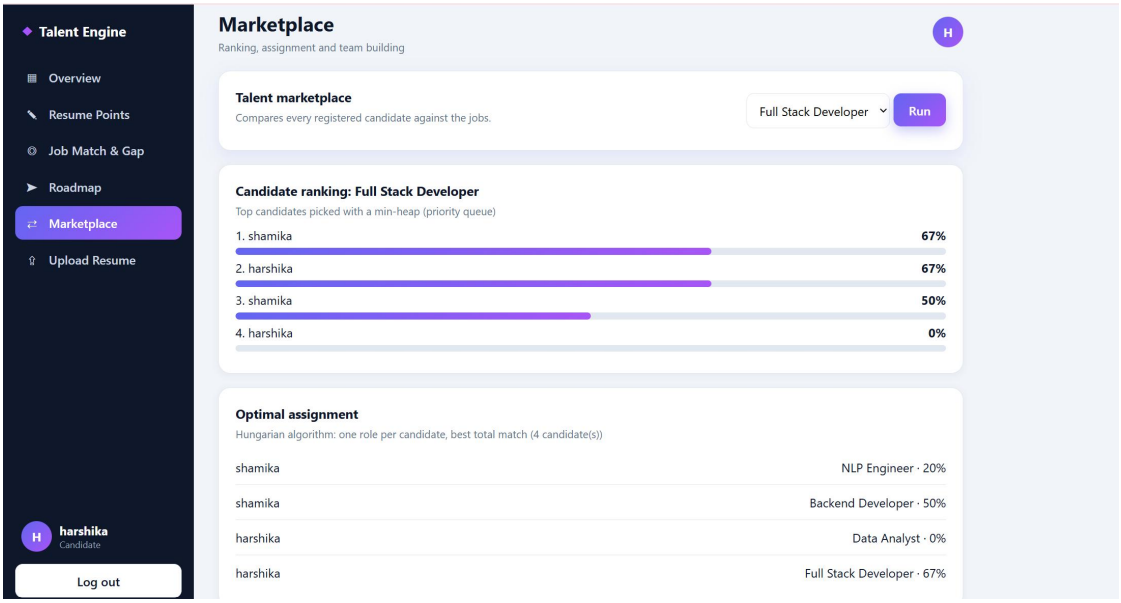
The system compares the candidate's extracted skills with job requirements and calculates the matching percentage. The Top-K recommendation mechanism using a Min-Heap provides the most relevant job opportunities to the candidate.

7.3 Skill-Gap Analysis and Learning Roadmap



The system identifies skills that are missing for a selected job and determines prerequisite relationships between skills. Graph algorithms, including **DFS and Kahn's Topological Sort**, are used to generate an ordered learning roadmap for the candidate.

7.4 Talent Marketplace and Candidate Assignment



The marketplace matches candidates with suitable job opportunities based on their skill coverage. The implementation also uses **Hungarian Algorithm-based assignment and Greedy Set Cover** techniques to support candidate selection and job assignment.

8. CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The **Resume–Job Matching & Talent-Marketplace Engine** successfully demonstrates the application of **Data Structures and Algorithms in a real-world recruitment system**. The system processes candidate resumes, extracts relevant skills, compares them with job requirements, and recommends suitable job opportunities. It also identifies skill gaps and generates learning roadmaps using graph-based algorithms. The implementation of **Map/HashMap, Set/HashSet, N-gram scanning, Min-Heap, DFS, Kahn's Topological Sort, Hungarian Algorithm, and Greedy Set Cover** demonstrates how DSA techniques can improve recruitment-related operations such as searching, matching, ranking, and assignment.

Overall, the project provides an integrated platform for connecting candidate skills with suitable employment opportunities while helping candidates understand the skills they need to develop for their desired career paths.

8.2 Future Scope

The project can be further enhanced in the following areas:

1. **Advanced AI-Based Resume Analysis**
   Integrate advanced NLP and machine-learning models for more accurate resume understanding and skill extraction.
2. **Improved Job Matching**
   Incorporate semantic similarity and contextual matching instead of relying primarily on extracted skills.
3. **Real-Time Job Data**
   Connect the system with external job platforms or APIs to provide continuously updated job opportunities.
4. **Personalized Career Recommendations**
   Improve career recommendations by considering candidate experience, interests, previous applications, and career goals.
5. **Enhanced Talent Marketplace**
   Add advanced recruiter–candidate interaction, application tracking, communication, and hiring workflows.
6. **Performance Optimization**
   Optimize the implemented DSA algorithms to handle significantly larger numbers of candidates, resumes, and job postings.

9. REFERENCES

- **Workday – Candidate Skills Match**

Workday Candidate Skills Match
Used as a reference for skill-based candidate–job matching and resume skill extraction.

- **Indeed – Smart Sourcing**

Indeed Smart Sourcing
Used as a reference for AI-assisted candidate sourcing and job matching.

- **Indeed – How to Find Candidates with Smart Sourcing**

Indeed Candidate Sourcing
Used for understanding skills, experience, and candidate matching approaches.

- **Indeed – Skills-First Hiring**

Indeed Skills-First Hiring
Used as a reference for skills-based candidate searching and matching.

- **Eightfold AI – AI-Powered Talent Matching**

Eightfold AI Talent Matching
Used as a reference for semantic talent matching, skill overlap, ranking, and explainable matching.

- **Eightfold AI – AI-Powered Screening**

Eightfold AI Candidate Screening
Used as a reference for skill-based candidate screening and candidate ranking.

- **Eightfold – Talent / Career Exchange**

Eightfold Talent
Used as a reference for resume-based job recommendations and talent marketplaces.

- **Eightfold AI – Skills Intelligence**

Eightfold Skills Intelligence
Used for understanding skills-based talent management and career development.

- **Workday – Prospects and Candidates**

Workday Prospects and Candidates
Used as a reference for resume parsing, skill suggestions, candidate comparison, and suggested jobs.

- **Indeed – Smart Sourcing Policies**

Indeed Smart Sourcing Policies
Used as a reference for matching candidates using relevant skills, experience, education, and qualifications.